

[EECS 304] Control I

Sheet (1) Laplace Transform

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Outline

1 Overview

- Laplace Transform
- Laplace Transform Pairs
- Laplace Transform Applications
- System Linearity

2 Problems

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- Problem 2-a
- Problem 2-d
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Laplace Transform

Definition

For a time function $f(t)$, Laplace transformation is obtained by evaluating

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st} dt$$

$$s = \sigma + j\omega$$

Laplace Transform Pairs

Time Function	Laplace transform
Unit step $u(t)$	$\frac{1}{s}$
Ramp function $tu(t)$	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$
$e^{-\alpha t}$	$\frac{1}{s + \alpha}$
$e^{-\alpha t} f(t)$	$F(s + \alpha)$
$\cos(at)$	$\frac{s}{s^2 + a^2}$
$f(t - t_0)$	$e^{-st_0} F(s)$
$\dot{f}(t)$	$sF(s) - f(0)$
$\ddot{f}(t)$	$s^2 F(s) - sf(0) - \dot{f}(0)$
$\int f(t) dt$	$\frac{F(s)}{s}$
$-tf(t)$	$\frac{d}{ds} F(s)$

Laplace Transform Applications

- Initial value Theorem

$$\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} sF(s)$$

- Final value Theorem [For **bounded f(t)**]

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

- Transfer Function [For **LTI** systems with **zero initial conditions**]

$$TF = \frac{O/P(s)}{I/P(s)} = \frac{Y(s)}{X(s)}$$

- Solving differential equations

System Linearity

- Testing linearity of systems

$$y = f(x)$$

- ▶ Homogeneity:

$$\text{if } x \rightarrow y$$

$$\therefore Kx \rightarrow Ky$$

- ▶ Additivity:

$$\text{if } x_1 \rightarrow y_1 \ \& \ x_2 \rightarrow y_2$$

$$\therefore x_1 + x_2 \rightarrow y_1 + y_2$$

- ▶ Superposition:

$$\text{if } K_1x_1 \rightarrow K_1y_1 \ \& \ K_2x_2 \rightarrow K_2y_2$$

$$\therefore K_1x_1 + K_2x_2 \rightarrow K_1y_1 + K_2y_2$$

Problem 1-b

Problem 1-b

Solve the following differential equation:

$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = \dot{x}(t) + 3x(t)$$

where $y(0) = 1$, $\dot{y}(0) = 0$ & $x(t) = e^{-4t}$

Problem 1-b

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Solution

Using Laplace Transform:

$$s^2 Y(s) - sy(0) - \dot{y}(0) + 3sY(s) - 3y(0) + 2Y(s) = sX(s) - x(0) + 3X(s)$$

$$x(0) = 1, \text{ \& } X(s) = \frac{1}{s+4}$$

$$\therefore Y(s)[s^2 + 3s + 2] = X(s)[s + 3] + [s + 2]$$

Problem 1-b

Solution

$$Y(s)[s^2 + 3s + 2] = X(s)[s + 3] + [s + 2]$$

$$\therefore Y(s) = \frac{s + 3}{(s + 1)(s + 2)}X(s) + \frac{s + 2}{(s + 1)(s + 2)}$$

$$\therefore Y(s) = \frac{s + 3}{(s + 1)(s + 2)(s + 4)} + \frac{1}{(s + 1)}$$

Using partial fractions:

$$\therefore Y(s) = \frac{A}{(s + 1)} + \frac{B}{(s + 2)} + \frac{C}{(s + 4)} + \frac{1}{(s + 1)}$$

$$\therefore A = \frac{s + 3}{(s + 2)(s + 4)} \Big|_{s=-1} = \frac{2}{3} \quad \therefore B = \frac{s + 3}{(s + 1)(s + 4)} \Big|_{s=-2} = \frac{-1}{2}$$

Problem 1-b

Solution

$$\therefore C = \frac{s+3}{(s+1)(s+2)} \Big|_{s=-4} = \frac{-1}{6}$$

$$\therefore Y(s) = \frac{\frac{2}{3}}{(s+1)} + \frac{\frac{-1}{2}}{(s+2)} + \frac{\frac{-1}{6}}{(s+4)} + \frac{1}{(s+1)}$$

$$\therefore Y(s) = \frac{\frac{5}{3}}{(s+1)} + \frac{\frac{-1}{2}}{(s+2)} + \frac{\frac{-1}{6}}{(s+4)}$$

Using Inverse Laplace:

$$y(t) = \frac{5}{3}e^{-t} - \frac{1}{2}e^{-2t} - \frac{1}{6}e^{-4t}, \quad t \geq 0$$

Problem 2-a

Problem 2-a

Test the linearity of the system described by the following i/p – o/p relation:

$$y(t) = au(t)$$

where 'a' is a constant

Problem 2-a

Solution

For input $u_1(t) = \alpha u(t)$:

$$\rightarrow y_1(t) = a\alpha u(t)$$

For input $u_2(t) = \beta u(t)$:

$$\rightarrow y_2(t) = a\beta u(t)$$

$\therefore$ System is homogenous

For input $u_3(t) = \alpha u(t) + \beta u(t)$:

$$\rightarrow y_3(t) = a(\alpha u(t) + \beta u(t))$$

$$\therefore y_3(t) = a\alpha u(t) + a\beta u(t) = y_1(t) + y_2(t)$$

$\therefore$ System is additive

Problem 2-a

Solution

$\therefore \text{Homogenous} + \text{Additive} \rightarrow \text{Linear System}$

Problem 2-d

Problem 2-d

Test the linearity of the system described by the following i/p – o/p relation:

$$\ddot{y}(t) + a\dot{y}(t) + y(t) = u(t), \quad y(0) = \dot{y}(0) = 0$$

Problem 2-d

Solution

Taking Laplace for both sides:

$$\therefore s^2 Y(s) - sy(0) - \dot{y}(0) + asY(s) - ay(0) + Y(s) = U(s)$$

$$\because y(0) = \dot{y}(0) = 0 \rightarrow Y(s)(s^2 + as + 1) = U(s)$$

$$\therefore Y(s) = \frac{U(s)}{s^2 + as + 1}$$

Convolution in time $\leftrightarrow$ Multiplication in S – plane

$$u(t) \circledast h(t) \leftrightarrow U(s) * H(s)$$

$$\therefore Y(s) = U(s) * \frac{1}{s^2 + as + 1} = U(s) * H(s)$$

Problem 2-d

Solution

$$\therefore y(t) = u(t) \circledast h(t)$$

$$\therefore h(t) = \ell^{-1}\left(\frac{1}{s^2 + as + 1}\right) = \ell^{-1}\left(\frac{1}{(s + \alpha_1)(s + \alpha_2)}\right)$$

$$\therefore h(t) = \ell^{-1}\left(\frac{\beta_1}{s + \alpha_1} + \frac{\beta_2}{s + \alpha_2}\right) = \beta_1 e^{-\alpha_1 t} + \beta_2 e^{-\alpha_2 t}$$

$$\therefore y(t) = u(t) \circledast h(t) = \int_0^t u(\tau) h(t - \tau) d\tau$$

$$\therefore y(t) = \int_0^t u(\tau) [\beta_1 e^{-\alpha_1(t-\tau)} + \beta_2 e^{-\alpha_2(t-\tau)}] d\tau$$

$$\therefore y(t) = \int_0^t u(\tau) \beta_1 e^{-\alpha_1(t-\tau)} d\tau + \int_0^t u(\tau) \beta_2 e^{-\alpha_2(t-\tau)} d\tau$$

Problem 2-d

Solution

$$\therefore y(t) = \int_0^t u(\tau)w(\tau)d\tau + \int_0^t u(\tau)z(\tau)d\tau$$

For input $u_1(t) = \alpha u(t)$:

$$\rightarrow y_1(t) = \alpha \int_0^t u(\tau)w(\tau)d\tau + \alpha \int_0^t u(\tau)z(\tau)d\tau = \alpha y(t)$$

For input $u_2(t) = \beta u(t)$:

$$\rightarrow y_2(t) = \beta \int_0^t u(\tau)w(\tau)d\tau + \beta \int_0^t u(\tau)z(\tau)d\tau = \beta y(t)$$

$\therefore$ System is homogenous

Problem 2-d

Solution

For input $u_3(t) = \alpha u(t) + \beta u(t)$:

$$\begin{aligned} \rightarrow y_3(t) &= \alpha \int_0^t u(\tau)w(\tau)d\tau + \beta \int_0^t u(\tau)w(\tau)d\tau \\ &+ \alpha \int_0^t u(\tau)z(\tau)d\tau + \beta \int_0^t u(\tau)z(\tau)d\tau \end{aligned}$$

$$\therefore y_3(t) = y_1(t) + y_2(t)$$

$\therefore$ System is additive

$\therefore$ Homogenous + Additive $\rightarrow$ Linear System

Problem 3-b

Problem 3-b

Find the Transfer Function of the following system:

$$\dot{y}(t) + y(t) = x(t - T)$$

Problem 3-b

Problem 3-b

Find the Transfer Function of the following system:

$$\dot{y}(t) + y(t) = x(t - T)$$

Solution

Using Laplace Transform with Z.I.C.:

$$\therefore sY(s) + Y(s) = e^{-Ts}X(s)$$

$$\therefore \frac{Y(s)}{X(s)} = \frac{e^{-Ts}}{s + 1}$$

Thank you!

Questions?

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